

INTERMEDIATE QUANTS – 4

Directions for questions 1 to 4: Select the correct alternative from the given choices.

Two alloys containing iron and copper in the ratio 3 : 5 and 1 : 3 are to be melted to form a 70 kg alloy having 30% iron. Find the quantity of each alloy that needs to be melted.

- ☐ a) 27 kg and 43 kg
- ☐ b) 36 kg and 34 kg
- ☐ c) 14 kg and 56 kg
- ☐ d) 28 kg and 42 kg

Solution:

Using alligation

$$\begin{array}{ccc} \frac{3}{8} & & \frac{1}{4} \\ & \searrow \quad \nearrow & \\ & \frac{3}{10} & \\ & \nearrow \quad \searrow & \\ 2 & & 3 \end{array}$$

Ratio of quantities to be mixed = 2 : 3 and total weight of the new alloy is 70kgs

∴ Respective quantities of alloys are 28kg and 42kg.

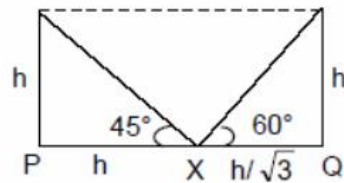
Choice (D)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

Poles P and Q of the same height are on either side of a 150 m road and are opposite to each other. From a point 'x' on the road, the angles of elevation of the tops of poles P and Q are 45° and 60° respectively. What is the minimum possible distance between 'x' and the foot of the pole Q?

- ☐ a) $75(\sqrt{3} - 1)$ m
- ☐ b) $75(\sqrt{3} + 1)$ m
- ☐ c) $150(\sqrt{3} - 1)$ m
- ☐ d) $75\sqrt{3}$ m

Solution:



Let the height of the poles be 'h' m.

$$h + \frac{h}{\sqrt{3}} = 150$$

$$h = \frac{150\sqrt{3}}{(\sqrt{3} + 1)}$$

The distance between x and foot of the pole Q is $\frac{h}{\sqrt{3}}$ i.e. $\frac{150}{\sqrt{3} + 1}$ or $75(\sqrt{3} - 1)$.

Choice (A)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

The angles of a quadrilateral ABCD in degrees are integers, and $\angle A \angle B \angle C < \angle D$. The sum of the two greater angles is 25% more than the sum of the two smaller angles. Find the greatest possible value of the measure of $\angle C$.

- ☐ a) 90°
- ☐ b) 89°
- ☐ c) 100°
- ☐ d) 99°

Solution:

If $\angle A + \angle B = 4x$, then $\angle C + \angle D = 5x$
i.e., $9x = 360^\circ$ or $x = 40^\circ$. $\therefore \angle C + \angle D = 200^\circ$.
The greatest possible value of $\angle C$ is 99°

Choice (D)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

Pipe P can pump 75% faster than pipe Q and pipe Q can pump 20% faster than pipe R. If P and R together can fill a tank in 7 hours, how much time does Q alone take to fill the tank?

- ☐ a) $17\frac{1}{13}$ hours
- ☐ b) $17\frac{1}{12}$ hours
- ☐ c) $18\frac{1}{13}$ hours
- ☐ d) $18\frac{1}{12}$ hours

Solution:

Let the rate at which Q fills the tank be $4k$.

$\therefore$ Rate at which P fills will be $7k$ and that of R will be $\frac{10}{3}k$

As P and R take 7 hours together, they'll fill $\left(7k + \frac{10k}{3}\right) \times 7$

To fill the same, Q will take $\frac{\frac{31k}{3} \times 7}{4k}$ i.e., $\frac{217}{12}$ hrs.

Choice (D)

Directions for question 5: Type in your answer in the input box provided below the question.

The LCM and HCF of three numbers are 9044 and 4 respectively. If two of the numbers are 68 and 28, find the least value of the third number.

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

Let the numbers be ha , hb , and hc , where h is the HCF of the three numbers and a , b and c are co-prime to each other. ($\because$ the third number has to be the least such number)

$$\therefore ha = 68, hb = 28$$

$$\text{As } h = 4, a = 17, b = 7$$

$$\text{LCM} = h \times a \times b \times c$$

$$9044 = 4 \times 17 \times 7 \times c$$

$$\Rightarrow c = 19$$

$$\therefore \text{The third number is } hc = 19 \times 4 = 76.$$

Ans : (76)

Directions for question 6: Select the correct alternative from the given choices.

If a , b , c and d are in GP, then $\log_a b$, $\log_b c$, $\log_c d$ _____.

- ☐ a) are in GP.
- ☐ b) can be obtained by adding a constant to terms in a GP.
- ☐ c) are in an HP.
- ☐ d) can be obtained by adding a constant to terms in an HP.

Solution:

If several terms are in a GP, their logs are in an AP. The ratio of consecutive terms of an AP are related to an HP. Consider the AP; a , $a + d$, $a + 2d$, $a + 3d$ and consider the ratios of consecutive terms $\frac{a+d}{a} = 1 + \frac{d}{a}$, $\frac{a+2d}{a+d} = 1 + \frac{d}{a+d}$, $\frac{a+3d}{a+2d} = 1 +$

$$\frac{d}{a+2d}, \dots$$

If we subtract 1 from these terms, we get an HP.

Now a , b , c , d are in GP. Therefore, $\log a$, $\log b$, $\log c$, $\log d$ are in AP and $\frac{\log b}{\log a}$, $\frac{\log c}{\log b}$, $\frac{\log d}{\log c}$ are obtained by adding 1 to the terms of an HP.

Choice (D)

Directions for question 7: Type in your answer in the input box provided in the question.

A boat made a round trip journey between two points 36 km apart on the banks of a river. The total time taken for the round trip was 7.5 hours. If the speed of the boat in still water is 10 kmph, the speed of the stream is

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

Let the speed of the stream be x kmph

$$\frac{36}{10+x} + \frac{36}{10-x} = 7.5$$

$$\frac{36(20)}{10^2 - x^2} = 7.5$$

$$x^2 = 4$$

$$x = 2 \quad (\because x > 0)$$

Ans : (2)

Directions for questions 8 to 11: Select the correct alternative from the given choices.

Find the positive values of x that satisfy the inequation

$$x^{\frac{2}{3}}$$

-

$$x^{\frac{1}{3}}$$

$$+ 46 \geq 102.$$

☐ a) $7 \leq x \leq 8$

☐ b) $x \geq 512$ (Correct Answer - ✓)

☐ c) $x \geq 0$

☐ d) $x \geq 8$

Solution:

$$x^{\frac{2}{3}} - x^{\frac{1}{3}} + 46 \geq 102$$

$$\text{i.e., } x^{\frac{2}{3}} - x^{\frac{1}{3}} - 56 \geq 0$$

$$\text{let } x^{\frac{1}{3}} = a$$

$$\Rightarrow a^2 - a - 56 \geq 0$$

$$(a + 7)(a - 8) \geq 0$$

$$a \leq -7 \text{ or } a \geq 8$$

As x is positive $x^{\frac{1}{3}} \leq -7$ is neglected

$$\Rightarrow a \geq 8 \Rightarrow x^{\frac{1}{3}} \geq 8$$

$$\Rightarrow x \geq 512$$

Choice (B)

Directions for questions 8 to 11: Select the correct alternative from the given choices.

At what time between 4 O' clock and 5 O' clock are the hands of a clock exactly at 90° with each other?

- ☐ a) $4 : 5 \frac{5}{11}$
- ☐ b) $4 : 6 \frac{5}{11}$
- ☐ c) $4 : 38 \frac{2}{11}$

☐ d) More than one of the above (Correct Answer - ✓)

Solution:

Between 4 O' clock and 5 O' clock, there will be two instances when the angle between the hands is 90° .

Times will be $\frac{30}{5.5}$ and $\frac{210}{5.5}$ i.e. $\frac{60}{11}$ minutes and $\frac{420}{11}$ minutes after 4 O' clock

where 5.5° per minute is the relative speed of the minute hand with respect to hour hand and 30° and 210° are the angles to be swept.

Choice (D)

Directions for questions 8 to 11: Select the correct alternative from the given choices.

$$X =$$

$$\sqrt{p^2 + q^2}$$

$$\text{and } Y =$$

$$\sqrt[3]{p^3 + q^3}$$

where p and q are positive numbers.

Consider the following statements:

$$\text{I. } X > Y$$

$$\text{II. } X \leq Y$$

Which of the following can be concluded?

- ☐ a) Only I is possible
- ☐ b) Only II is possible
- ☐ c) Both I and II are possible
- ☐ d) $X > Y$ or $X < Y$

Solution:

$$\begin{aligned} X^6 &= (p^2 + q^2)^3 \text{ and } Y^6 = (p^3 + q^3)^2 \\ X^6 - Y^6 &= p^6 + q^6 + 3p^2q^2(p^2 + q^2) - (p^6 + q^6 + 2p^3q^3) \\ &= p^2q^2(3p^2 + 3q^2 - 2pq) \\ &= p^2q^2(3(p - q)^2 + 4pq) \\ 3(p - q)^2 + 4pq &\text{ is always positive} \\ \therefore X^6 - Y^6 &\text{ is always positive i.e. } X^6 > Y^6 \\ \therefore X &> Y. \end{aligned}$$

Choice (A)

Directions for questions 8 to 11: Select the correct alternative from the given choices.

The sum of three multiples of 5 – $5a$, $5b$, and $5c$, (where $a \leq b \leq c$) is 125. If the sum of $5a$ and $5b$ is greater than or equal to one and a half times $5c$, how many such triplets of positive integers are there?

- ☐ a) 6
- ☐ b) 3
- ☐ c) 4
- ☐ d) 5

Solution:

The 3 multiples of 5 are $5a$, $5b$, $5c$. Therefore, $a + b + c = 25$. (where $a \leq b \leq c$).

Also, $a + b \geq \frac{3}{2}c$

$$\Rightarrow a + b + c \geq \frac{5c}{2} \Rightarrow c \leq \frac{2}{5}(a + b + c),$$

i.e., $c \leq 10$.

$\therefore$ The possible values of (a, b, c) are $(5, 10, 10)$ $(6, 9, 10)$, $(7, 8, 10)$ $(7, 9, 9)$ $(8, 8, 9)$.
There are 5 such values. Choice (D)

Directions for question 12: Type in your answer in the input box provided below the question.

A set has 5 elements. In how many ways can the elements of the set be divided into two groups?

Solution:

Method 1:

Each group must have at least one element

Either the number of elements in the two groups must be 1 and 4 or 2 and 3.

If one of the groups has one element, E(say), and the other has four elements, E can be any one of the five elements. Number of ways to divide the elements of X into two groups (N_1) = 5

If one of the groups has two elements, A and B (say), the other has three elements. A and B can be any two of the five elements. Number of ways to divide the elements of X into two groups (N_2) = 5C_2 .

Number of ways of dividing the elements of X into two groups = $N_1 + N_2 = 15$.

Method 2:

Each element of X can go into any of the two groups. Number of ways of division = $2^5 = 32$ but, we cannot have any of the groups empty.

Number of ways of division (Such that no group is empty) (N) = $32 - 2 = 30$

Suppose there exists a division resulting in a elements in group 1 and 4 - a elements in group 2. Then there also exists a division resulting in a elements in group 2 and 5 - a elements in group 1. These divisions are not different.

Number of ways of division = $\frac{N}{2} = 15$

Ans : (15)

Directions for question 13: Select the correct alternative from the given choices.

An insect is sitting at one of the corners of a cubical room. The maximum distance that it has to fly to reach any point in the room is

$$10\sqrt{3}$$

m. Find the edge of the room.

- ☐ a) $5\sqrt{3}$ m
- ☐ b) 5 m
- ☐ c) 10 m
- ☐ d) $\frac{10\sqrt{3}}{3}$ m

Solution:

Let the edge of the cubical room be a

The farthest point from the corner (say) where the insect is sitting is the corner diagonally opposite to C.

Maximum distance that the insect has to fly to reach a point on the room = $a\sqrt{3}$

$$a\sqrt{3} = 10\sqrt{3}$$

$$a = 10$$

Choice (C)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

$$x^2 + y^2 = 0.9$$

$$|x - y| = 0.8$$

$$|x| + |y| = c$$

$$\sqrt{29}.$$

Find c.

Solution:

$$\begin{aligned} |x - y| &= 0.8 \\ \Rightarrow (x - y)^2 &= 0.64 \\ \Rightarrow x^2 + y^2 - 2xy &= 0.64 \\ \Rightarrow 0.90 - 0.64 &= 2xy \\ \Rightarrow 2xy &= 0.26 \\ x^2 + y^2 &= 0.9 \\ \Rightarrow x^2 + y^2 + 2xy &= 0.9 + 2xy \\ \Rightarrow (|x| + |y|)^2 &= 0.90 + 0.26 = 1.16 \quad (\because 2xy = 2|xy| = |2xy|) \\ |x| + |y| &= \frac{\sqrt{1.16}}{5} = \frac{\sqrt{29}}{5} = c \sqrt{29}. \end{aligned}$$

Ans : (0.2)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

Every month, Sara saves 40% of her monthly salary. Due to an increase in costs, her expenditure this month increased by 15%. Hence, she is able to save only Rs.18,600. What is her monthly salary (in Rs.)?

Solution:

$$\begin{aligned} \text{Let her monthly salary be ₹} x \\ \text{She saves 40\% i.e. } 0.4x \\ \therefore \text{ She spends } 0.6x \\ \text{Expenditure got increased by 15\% and it therefore became } 0.6x \times 1.15 = 0.69x \\ \therefore \text{ Savings} = x - 0.69x = 0.31x \text{ and } 0.31x = 18600 \\ \Rightarrow x = 60,000. \end{aligned}$$

Ans : (60000)